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Course/Year/Sec: BSIT 3B

Individual Activity4: "Design Your Own System Using UML"

Part 1: System Idea & UML Diagram Recall

Task 1: List and Describe UML Diagrams

1. **Use Case Diagram:** A behavioral diagram that identifies the key actors involved in a system and the specific tasks (use cases) they can perform, defining the system's requirements and boundary.
2. **Activity Diagram:** A flowchart-style diagram that details the dynamic step-by-step logic of a specific process, such as the flow of placing an order, including sequential activities and decisions.
3. **Class Diagram:** A structural diagram that describes the internal static structure of a system by showing the different classes (objects), their attributes, methods, and the relationships between them (like inheritance or composition).
4. **Sequence Diagram:** An interaction diagram that illustrates how objects in a system collaborate with each other over time, specifically showing the order of messages exchanged between objects to complete a particular scenario.

Task 2: System Idea Description

System Title

"Online Floral Management System"

Short Description:

The Online Floral Management System is a web-based and mobile application designed to streamline the operation of a modern flower shop. It serves as an intuitive e-commerce platform where customers can easily browse an extensive catalog of floral arrangements and use interactive tools to fully customize their bouquets. The system handles secure payment processing and order fulfillment, while also providing robust back-end tools for shop owners (admins) to efficiently manage product inventory and track essential sales data through detailed reports. This creates a seamless, digital experience for both the consumer and the business owner.

Main Features:

1. **Interactive Floral Catalog & Interactive Customizer:** An online storefront with search functionality, integrated with a drag-and-drop tool for customers to design bouquets by flower type and color.

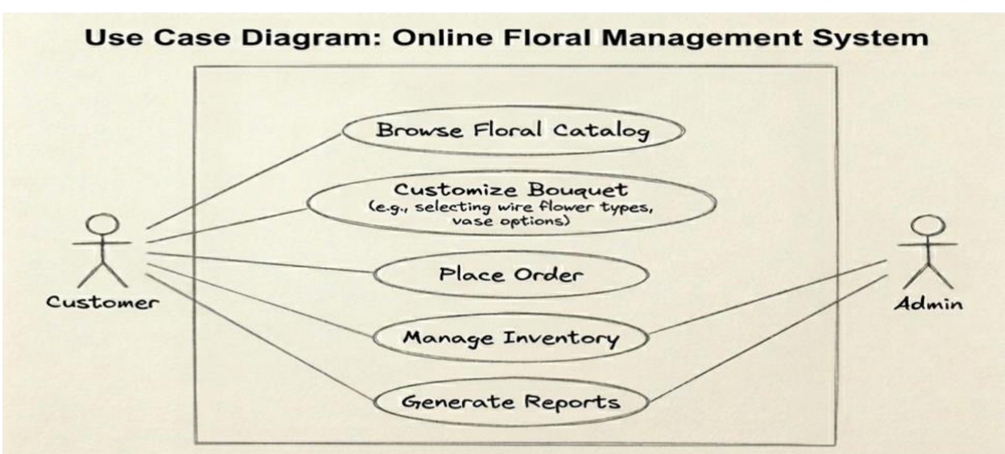
2. **Order Placement & Secure Payment Processing:** A standard shopping cart system and checkout process that integrates with secure payment gateways.

3. **Real-time Inventory Management:** A system for the Admin to update stock levels, set low-stock alerts for flowers and vases, and receive new arrivals.

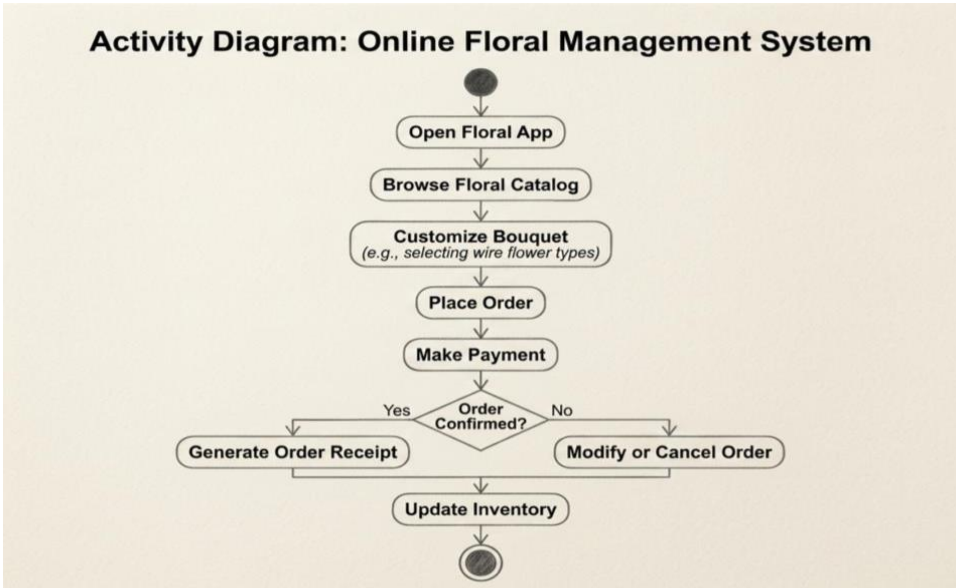
4. **Sales Reporting and Customer Analytics:** Tools to generate automated daily, weekly, and monthly sales reports, as well as track customer ordering patterns to optimize marketing efforts.

Part 2: UML Diagram Creation

Use Case Diagram



Activity Diagram – "Process Flow for Ordering a Bouquet"



Part 3: SDLC Application

This section outlines how the specific phases of the Software Development Life Cycle apply to the creation of the Online Floral Management System.

1. **Planning Phase:**

During this phase, we would define the system's primary goal, such as increasing shop sales by 20%, and determine that an e-commerce platform is the best solution. Key milestones and resource requirements, such as hiring developers or selecting server hosting, would be established.

2. **Design Phase:**

This is the stage where the provided UML diagrams (the Use Case Diagram and Activity Diagram) are developed to map user interactions and process logic. Architectural designs and user interface wireframes would also be drafted to guide developers.

3. **Implementation (Development) Phase:**

In this phase, programmers translate the design into functional code, building the front-end catalog using HTML/JavaScript and the back-end inventory logic using a database. This is when the actual floral interactive customizer tool and payment gateway are integrated.

4. **Testing Phase:**

The testing team performs comprehensive QA checks, such as verifying that inventory count decreases correctly when an order is placed (a logic test) and testing with actual credit cards to confirm that payment processing works securely without failure.

Part 4: Reflection

What did you learn from creating UML diagrams?

Creating UML diagrams helped me break down a complex system into manageable visual components, showing me that planning user requirements and process flows before coding is essential for clarity.

Why is planning (SDLC) important before building a system?

A strong, disciplined planning phase using the SDLC helps teams avoid costly scope creep, identifies technical challenges early, and ensures the entire development team is aligned with the final business goals.